



Poverty Prediction and Regional Forecasting in Armenia

Machine Learning on ILCS 2015 Household Survey Data
and ArmStat Regional Panel (2016–2022)

Author Hovo Sukiasyan

Supervisor Aleksandr Hayrapetyan

Program Chair Hayk Nersisyan

Institution American University of Armenia

Yerevan, Armenia

Abstract

Poverty measurement in Armenia relies heavily on periodic household surveys and annual regional estimates, leaving a gap between data collection cycles and timely policy decisions. This project addresses that gap in two strands: supervised machine learning on ILCS 2015 to predict household income, and regional time-series forecasting on ArmStat statistics for Armenia’s eleven marzes (2016–2022).

The most informative regional signal is available at **annual** (survey-based) resolution; neural sequence models benchmarked alongside Lag-1, Ridge autoregressive, and ARIMA baselines systematically **underperform** Lag-1 on yearly poverty rate because each marz supplies only seven true annual training points (§5). Auxiliary **monthly** and **daily** views are obtained by forward-filling annual poverty rates; at those granularities goodness-of-fit is almost mechanically inflated because very little genuine movement occurs within survey years (§5).

Out-of-sample **credibility** is established by a strict **2022 holdout backtest**: models are trained exclusively on 2016–2021 data and scored against published 2022 rates. Lag-1 attains the best regional mean absolute error (5.07 percentage points); a 50/50 Lag-1/Ridge ensemble attains nearly zero population-weighted national error (0.02 percentage points).

Results and 2023–2026 projections are communicated through an interactive Next.js dashboard. This document is the project’s primary technical reference on data, methodology, architecture, and findings.

Contents

1	Introduction	4
1.1	System Overview	4
1.2	Objectives	4
1.3	Scope	5
2	Methodology	6
2.1	Literature Context and Motivation	6
2.2	Data Sources	6
2.3	Feature Engineering and Preprocessing	7
2.4	Validation Strategy	7
3	System Architecture	8
3.1	Overview	8
3.2	Technology Stack	9
4	Data Description and Schema	10
4.1	ILCS 2015 Dataset	10
4.2	ArmStat Regional Panel	11
4.3	Key Descriptive Statistics	11
4.4	Seasonal Income Variation	12
5	Model Development and Results	13
5.1	Household Income Models	13
5.2	Feature Importance	13
5.3	Bayesian Hyperparameter Optimization	14
5.4	Time-Series Analysis — Classical Models	16
5.5	Time-Series Analysis — Neural Network Models	17
5.6	2022 Forecast Validation	18
6	Application and Functionality	22
6.1	Homepage	22
6.2	Regional Explorer	23
6.3	Household Income Model	25
6.4	Feature Importance Page	26
6.5	Forecasting Diagnostics	27
6.6	2022 Forecast Validation	28
6.7	Future Projection 2023–2026	30
7	Challenges Encountered	32
7.1	Data Quality and Panel Size	32
7.2	Anomalous Regional Observations	32
7.3	Interview Month Recovery	32
7.4	Web Application Data Architecture	32
7.5	Model Evidence Communication	32

8	Future Work	33
8.1	Larger Regional Panels	33
8.2	Improved Household Income Models	33
8.3	Scenario Forecasting	33
8.4	Confidence Intervals on Projections	33
8.5	ILCS Longitudinal Analysis	33
9	Data and Code Availability	35
10	Conclusion	36
	References	37

1 Introduction

1.1 System Overview

The project is a full-stack data science system for predicting and forecasting household poverty in Armenia. It combines two data assets — a nationally representative household survey (ILCS 2015) [6] and a seven-year regional panel from the Armenian Statistical Committee (ArmStat) [5] — with a layered modelling pipeline and an interactive web interface.

At the micro level, cross-sectional ML models on ILCS 2015 identify which household characteristics (income source diversification, consumption expenditures, asset ownership) are most predictive of total household income, in the spirit of proxy-means assessment and machine-learning welfare targeting [1, 2]. At the macro level, time-series models on the ArmStat panel [5] estimate the future trajectory of regional poverty rates and a composite socioeconomic stress index, with credibility established through a 2022 holdout backtest trained entirely on 2016–2021 data.

The research findings and forecasts are exposed through a Next.js web application [10] (<https://armenia-poverty.vercel.app>) that allows users to explore regional trends interactively, inspect model performance metrics, review the 2022 validation evidence, and examine 2023–2026 poverty projections. A PostgreSQL (Neon) database [11] supports the dynamic data features, while validation results and forecast outputs are served as static CSV files for reproducibility. Data artefacts, scripts, and deployment details are summarised in Section 9.

1.2 Objectives

The primary objectives of this project are as follows.

1. Identify the household-level features most predictive of total income in Armenia using supervised machine learning on ILCS 2015 data, and quantify model performance through rigorous train/test evaluation with Bayesian hyperparameter optimization.
2. Construct and validate autoregressive and neural time-series models that forecast regional poverty rates across Armenia’s 11 marzes, using a strict held-out 2022 backtest for credibility assessment.
3. Build and deploy a public-facing web dashboard that communicates results, spatial patterns, and forecasts in an accessible, interactive format suitable for researchers and policy practitioners.
4. Document the complete data pipeline, modelling workflow, and web system architecture so that the project can be reproduced and extended.

A secondary objective is methodological: to honestly assess the limitations of poverty prediction with small regional panels and cross-sectional survey data, and to communicate uncertainty clearly rather than overstate model capability.

1.3 Scope

The project covers the following scope.

Data: (i) ILCS 2015 public-use household microdata — 5,184 households, 29 research columns after cleaning; (ii) ArmStat regional statistics — 11 marzes, 7 annual observations per marz (2016–2022), disaggregated to monthly frequency for time-series modelling.

Models: Five supervised regressors (Random Forest, Gradient Boosting, Extra Trees, Ridge, Lasso) with and without Bayesian hyperparameter optimization; four neural sequence models (GRU, BiLSTM, Transformer, TCN) trained at yearly, monthly, and daily granularity; and classical baselines (Lag-1, Ridge AR, ARIMA).

Web application: A Next.js/React web app hosted on Vercel, with pages for the homepage, regional explorer, regional trends, regional rankings, household model, feature importance, forecasting diagnostics, 2022 validation, and future projections.

Out of scope: Real-time data ingestion, mobile clients, causal inference, and individual-level poverty status classification (as opposed to income regression).

2 Methodology

The project follows an applied quantitative research design combining secondary data analysis, supervised machine learning, and time-series forecasting. All models are evaluated against held-out test data; no model is assessed only on training performance.

2.1 Literature Context and Motivation

Poverty measurement in low- and middle-income countries often relies on proxy means tests (PMT): scores based on observable characteristics when direct income measurement is imperfect or costly [1]. Machine-learning extensions of welfare scoring have demonstrated that ensemble trees and gradient boosting can outperform simple linear scores out of sample on survey data [2]. Absolute predictive power nonetheless remains modest in many applications, which is consistent with an R^2 of order 0.35 for household income on a single ILCS cross-section in this project.

For regional poverty forecasting, autoregressive models remain competitive with neural alternatives on short panels [3]. Armenia’s annual regional data (7 observations per marz) is at the lower bound for stable neural network training, and the results confirm that Lag-1 and Ridge AR outperform GRU and LSTM on the annual benchmark, while neural models are competitive only at monthly granularity where the data is synthetically interpolated.

2.2 Data Sources

ILCS 2015 Household Survey

The Integrated Living Conditions Survey (ILCS) is Armenia’s primary household welfare instrument, conducted by ArmStat. The 2015 wave covers 5,184 households across all marzes and urban/rural strata. After extraction, subsetting, and cleaning, 29 features are retained for ML modelling. These include consumption aggregates (food, services, goods), asset indicators (car ownership, computer, dwelling condition), household demographics (size, age structure), financial behaviour (debt, remittances, money transfers), and the interview month — recovered during preprocessing and critical for controlling seasonal income variation.

ArmStat Regional Panel

Annual statistics for 11 Armenian marzes covering 2016–2022 were sourced from the Armenian Statistical Committee. Variables include the official poverty rate (%), crime rate per 100,000 population, hospital count per 100,000, hospital beds per 10,000, and total population. The panel is augmented to monthly frequency via forward-filling and stored as a single augmented monthly table (924 rows: 11 marzes $\times$ 84 months).

A composite **Stress Index** is defined by applying standardisation twice: first to each input series, then to their signed sum. Let $z(\cdot)$ denote mapping a series to zero mean and unit variance (standardisation). Then

$$\text{stress index} = z\left(z(\text{poverty rate}) + z(\text{crime rate per } 100k) - z(\text{hospitals per } 100k)\right). \quad (1)$$

High values indicate co-occurring high poverty and high crime together with relatively low hospital capacity per 100,000 population.

2.3 Feature Engineering and Preprocessing

Monetary variables in ILCS 2015 span four orders of magnitude (50 AMD to 94.5 million AMD), so rescaling every column to zero mean and unit variance in one pass would overreact to heavy tails. A mixed strategy is used instead: eleven expenditure and income aggregates with pronounced right tails are mapped through a $\log(1 + x)$ transform first, then each is shifted to zero mean with unit variance; eight noisy continuous fields are scaled using medians and interquartile ranges so outliers do not dominate; the remaining nine binary and ordinal indicators are squeezed into the $[0, 1]$ interval so they sit on comparable footing with bounded inputs without assuming Gaussian shocks. The interview month (originally absent from the subset) is recovered by exploiting a deterministic relationship between household identifiers and the survey wave structure, yielding a perfectly balanced distribution (432 households per month, $n = 12$ months).

Separately from the ILCS preprocessing above, regional time-series work feeds neural sequence models twelve-month histories per marz whenever LSTM-family layers are invoked. For ridge autoregression fit on annual ArmStat aggregates, dependence enters through one-year and two-year lags (**lag-1**, **lag-2**) stacked alongside the contemporaneous regressors documented earlier.

2.4 Validation Strategy

Household models: Standard 80/20 stratified train/test split on ILCS 2015. Bayesian optimization (Optuna, 50 trials, 5-fold CV objective) is performed on the training fold only; the test fold is touched once for final reporting.

Time-series models: Temporal train/validation/test split at ≤ 2019 / 2020 / ≥ 2021 for within-sample evaluation. The primary evidence of forecast credibility is a **held-out 2022 backtest**: all models are retrained on 2016–2021 data and evaluated against actual 2022 regional poverty rates withheld during training. A rolling validation experiment further tests the same models on 2020, 2021, and 2022 as successive holdout years.

3 System Architecture

3.1 Overview

The system is organized into three layers: a data and modelling pipeline, a PostgreSQL data store, and a Next.js web application.

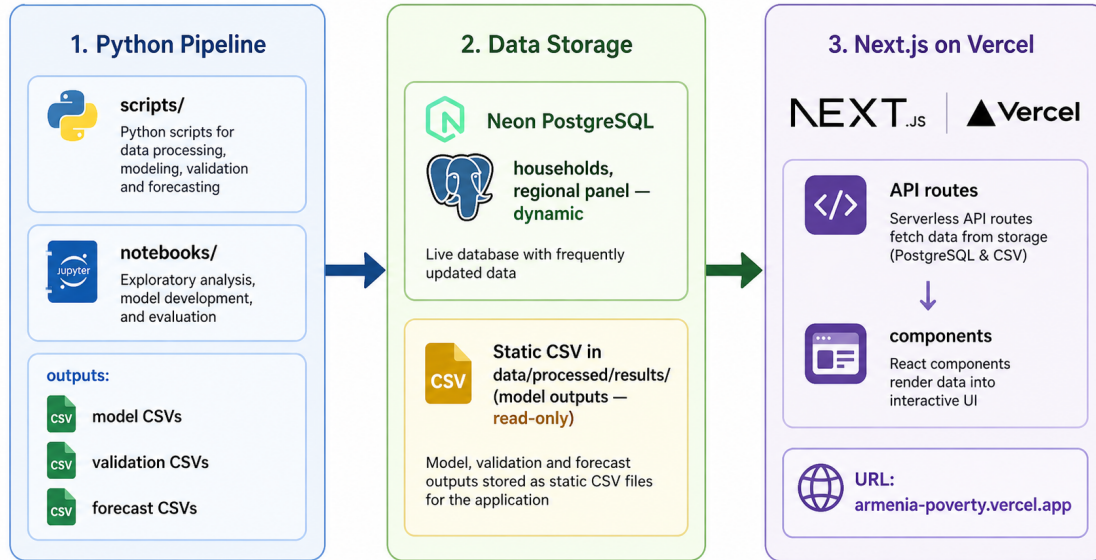


Figure 1: High-level system architecture showing the three-layer design: Python data and modelling pipeline (scripts and notebooks), hybrid data storage (Neon PostgreSQL for dynamic household and regional data; static CSV for model outputs), and Next.js web application deployed on Vercel.

The **data pipeline** runs entirely in Python: versioned runner scripts handle extraction, cleaning, feature engineering, ML training, and time-series analysis, complemented by Jupyter notebooks for exploration and prototyping. Key pipeline outputs — model metrics, forecast CSVs, and validation results — are committed to the repository and consumed by the web layer.

The **data store** uses two mechanisms: a PostgreSQL database hosted on Neon for dynamic, query-driven features (choropleth maps, distribution charts, scatter plots, household explorer); and static CSV files read at request time for model metrics, validation results, and forecasts. This hybrid approach eliminates unnecessary database queries for data that changes only when a new model run is committed.

The **web application** is a Next.js 14 app with React server components and selective client-side bundles only where hooks or browser APIs require them (Next.js client components). Next.js API routes serve JSON to those islands when server rendering alone is

insufficient. The app is deployed on Vercel with automatic redeployment whenever changes merge into the main deployment branch on Git.

3.2 Technology Stack

3.2.1 *Data and Modelling*

- **Python 3.11** — core language for data processing and modelling.
- **pandas, NumPy** — data handling and feature preparation.
- **scikit-learn** — machine learning models and evaluation.
- **Optuna** — hyperparameter tuning.
- **PyTorch** — deep learning models for time-series forecasting.
- **statsmodels** — statistical analysis and baseline models.

3.2.2 *Web Application*

- **Next.js** — full-stack framework for building the application.
- **Recharts** — data visualization.
- **Tailwind CSS** — styling.
- **SWR** — data fetching.

3.2.3 *Data Storage and Deployment*

- **PostgreSQL (Neon)** — database for storing data.
- **Vercel** — deployment platform.
- **Static CSV files** — storage of model outputs and results.

4 Data Description and Schema

4.1 ILCS 2015 Dataset

After cleaning and subsetting, the ILCS 2015 machine learning dataset contains 5,184 rows and 29 columns. The target variable is total annual household income (measured in AMD). The remaining predictors span five substantive groups:

1. **Consumption expenditures:** food purchases, purchases of goods and services, and overlapping “goods vs. services” totals.
2. **Financial behaviour:** outstanding family debt; money borrowed, lent, received domestically; and cash sent abroad in the preceding year.
3. **Asset and housing:** car ownership, computer ownership, perceptions of dwelling quality, and whether the household reports building new housing.
4. **Household demographics:** roster size plus the recorded count of income sources tied to labor, transfers, pensions, etc.
5. **Survey metadata:** recovered interview month indicators (1–12), humanitarian assistance receipt, and the subjective monetary minimum the household cites as survivable (*AMD-3 prompt*).

ILCS 2015 ML Dataset — Representative Fields

Household identifier

Total household income (*target, AMD*)

Count of household income categories

Purchases of services and mixed goods

Food purchases totals

Overlapping purchases of goods and services

Car ownership indicators (*coded 1=Yes, 2=No*)

Computer ownership indicators (*coded 1=Yes, 2=No*)

Family debt exposures

Household roster size

Reported dwelling quality

AMD-3 subjective minimum income

Recovered calendar month (1–12) of interview

4.2 ArmStat Regional Panel

The regional panel covers Armenia’s 11 administrative regions (marzes) across 84 months (January 2016 – December 2022). The yearly resolution data is the authoritative signal; monthly values are forward-filled from annual estimates for use in LSTM/GRU training.

ArmStat Regional Panel — Columns

Administrative region (marz)

Month

Regional poverty incidence (*%*, *ArmStat headline*)

Reported crimes per 10⁵ residents

Operational hospitals per 10⁵ residents

Hospital beds per 10⁴ residents

Headline population totals

Derived stress composite (Equation (1))

4.3 Key Descriptive Statistics

The income distribution in ILCS 2015 is highly right-skewed (min: 50 AMD, max: 3,405,000 AMD, $CV \approx 0.98$), confirming the need for log-scale visualizations and hybrid preprocessing.

Household characteristic	Affirmative (code=1)	Negative (code=2)
Has personal computer	3,082 (59.5%)	2,102 (40.5%)
Household possesses a motor vehicle	1,618 (31.2%)	3,566 (68.8%)
Received cash or gifts from others	2,461 (47.5%)	2,723 (52.5%)
Borrowed money in preceding 12 months	2,103 (40.6%)	3,081 (59.4%)
Received humanitarian assistance	145 (2.8%)	5,039 (97.2%)

Table 1: Binary feature distribution in ILCS 2015 (n=5,184 households)

Readers should beware a coding convention on these dichotomous questions: correlations with income flip sign because affirmative answers are labelled with the lower numeric code rather than the higher one. Households ticking “yes” to vehicle ownership or PCs sit on code 1, while “no” answers carry code 2, so larger numeric values actually flag asset-poor homes ($r \approx -0.32$ and -0.29 if you forget to swap the interpretation). Charts in the exploratory notebook and mirrored dashboard annotate the legends explicitly because otherwise the polarity looks “backwards” to casual analysts.

4.4 Seasonal Income Variation

The reconstructed interview-season indicator reveals systematic seasonality: households surveyed in January–March report approximately 20–25 % lower mean income than those surveyed in November–December (167K AMD vs. 202K AMD). This is consistent with agricultural income cycles and reduced off-season economic activity in rural Armenia. Because the ILCS is a single cross-section, seasonality cannot be absorbed with macro time trends; calendar month ranks are included as ordinal controls alongside other household regressors.

5 Model Development and Results

5.1 Household Income Models

Five supervised regression models are trained on ILCS 2015 to predict observed annual household income. The training set contains 4,147 households (80%); the test set contains 1,037 households (20%). Table 2 collects their out-of-sample scores on that 20 % holdout (R^2 , mean absolute error, and root mean squared error in AMD). Gradient boosting after Bayesian tuning emerges as the strongest specification on every column of the comparison.

Model	R^2	MAE (AMD)	RMSE (AMD)
Gradient Boosting (baseline)	0.3417	84,477	164,624
Random Forest	0.3339	85,475	165,603
Ridge Regression	0.3153	91,183	167,894
Lasso Regression	0.3154	91,186	167,879
Extra Trees	0.3146	87,681	167,979
GBM + Bayesian Opt	0.3563	82,152	162,784

Table 2: Household income model performance on held-out test set (1,037 households). GBM with Bayesian optimization is the best overall model.

5.2 Feature Importance

Feature importance rankings are consistent across all five models. *Household income source count* — the number of distinct income streams a household reports — is the single strongest predictor (correlation with income: $r = 0.45$; largest share in tuned gradient boosting: 0.32). The top five features across random forests and boosted trees are:

1. *Household income source count* — diversification is the clearest poverty signal. Households with only one income source (typically pension or a single wage) are substantially more vulnerable.
2. *Purchases of services and goods totals* — expenditures reflect liquidity better than dormant assets across this wave.
3. *Subjective survivable income (AMD-3)* — respondent-reported thresholds that track latent consumption floors.
4. *Overlapping purchases of goods and services* — confirms that spending composites jointly explain welfare rather than lone drivers.

5. *Food purchases totals* — captures subsistence shocks more directly than balance-sheet proxies.

Asset proxies (dwellings, automobiles, PCs) plateau near 2–4% absolute importance scores, signalling that contemporaneous expenditures dominate survey-based welfare contrasts for 2015.

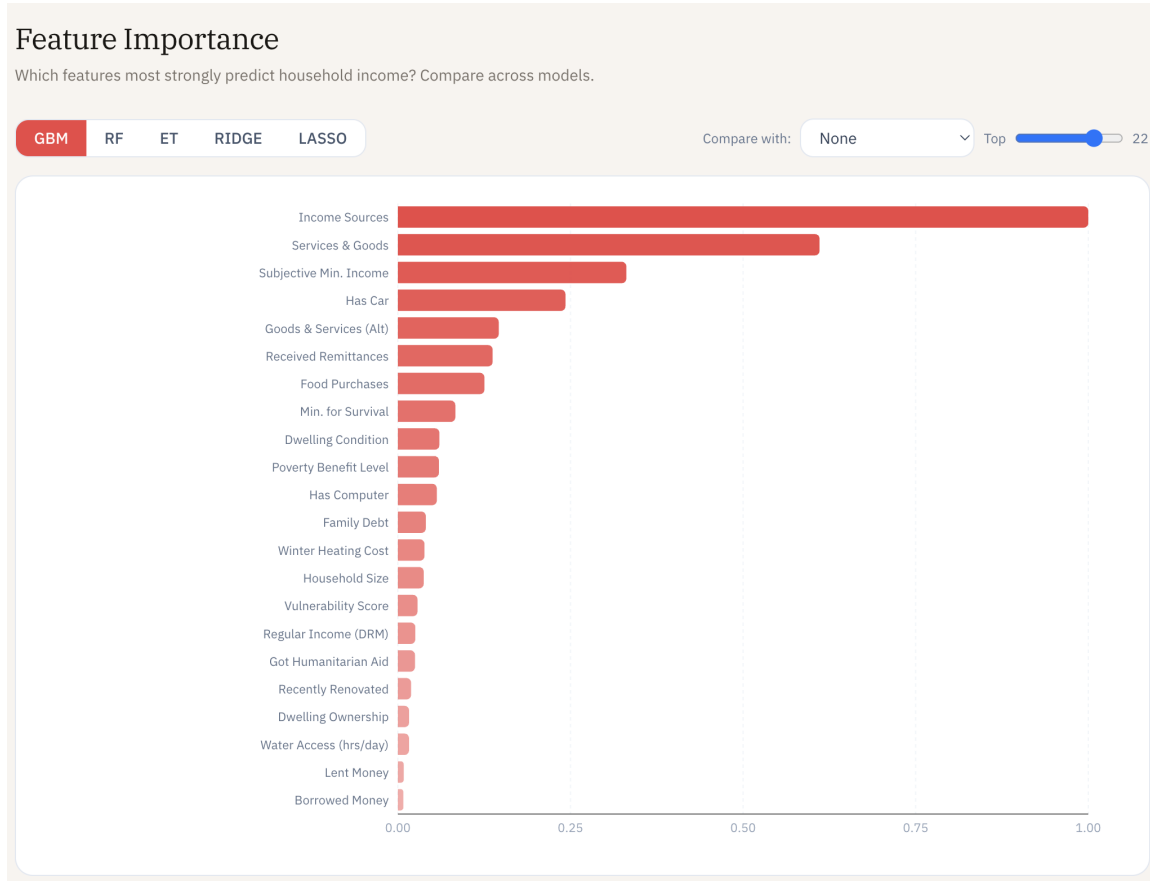


Figure 2: Normalized feature importance across all five supervised models (grouped bar chart). Income source diversification leads every architecture; expenditures dominate ranks while vehicle and electronics indicators trail because of asymmetric survey coding.

5.3 Bayesian Hyperparameter Optimization

Optuna [4] is used to optimize GBM, Random Forest, and Extra Trees over 50 trials each, with 5-fold cross-validation R^2 as the objective. Table 3 contrasts each model’s pre-tuning score, the best cross-validated R^2 returned by search, and the final test R^2 that already appeared for GBM in Table 2.

Model	Baseline R^2	CV R^2 (Bayes)	Test R^2
Random Forest	0.3339	0.4267	0.3398
Gradient Boosting	0.3417	0.4344	0.3563
Extra Trees	0.3146	0.4163	0.3316

Table 3: Bayesian optimization results. GBM benefits most from tuning. The CV–Test gap (0.43 vs. 0.36) reflects mild overfitting to training folds.

The best-performing boosted specification limits depth to six splits, fits 170 small trees sequentially with a gradual 3.1 % step size, and row-subsamples about 85.8 % of the training matrix each iteration—a conservative regime that slows overfitting yet still absorbs interactions among overlapping expenditure fields.

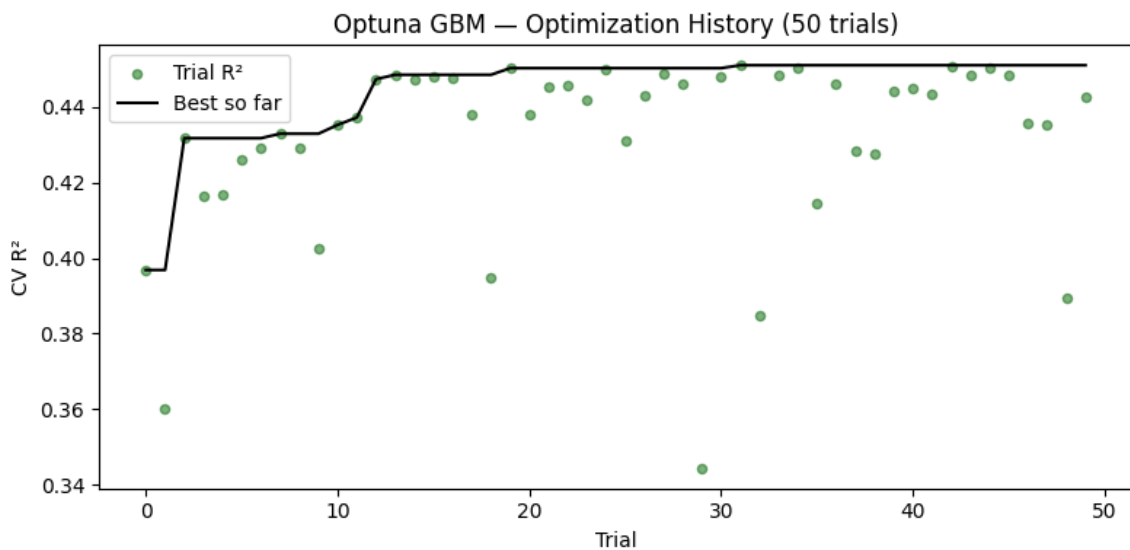


Figure 3: Optuna Bayesian optimization history for Gradient Boosting: 5-fold CV R^2 across 50 trials using the Tree-structured Parzen Estimator (TPE) sampler. The best trial reaches CV $R^2 = 0.4344$, corresponding to the configuration reported in the text.

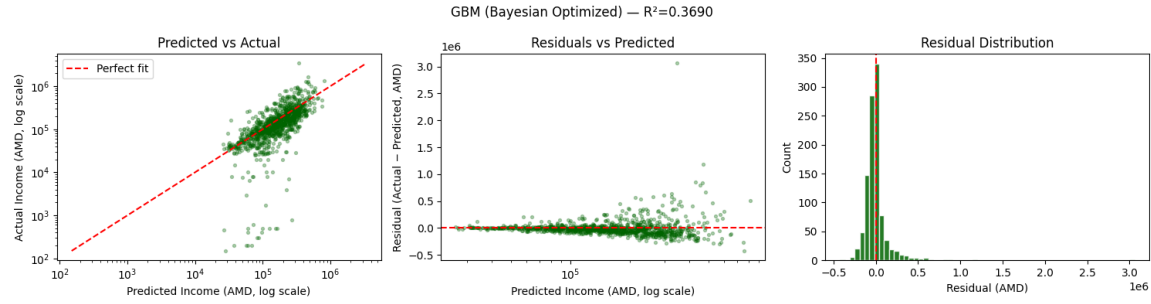


Figure 4: Predicted vs. actual household income (AMD) for the tuned Gradient Boosting model on the held-out test set (1,037 households, log–log scale). Systematic underestimation at the high end reflects the right-skewed income distribution; the central mass is well-calibrated.

5.4 Time-Series Analysis — Classical Models

A standalone Python benchmarking pass evaluates classical time-series models on the ArmStat regional panel at three temporal granularities. The train/validation/test split uses ≤ 2019 / 2020 / ≥ 2021 .

Model	Target	Frequency	R^2	MAE
Lag-1 Baseline	Poverty rate	yearly	0.728	5.11 pp
Lag-1 Baseline	Stress index	yearly	0.764	0.433
Ridge lags [1,2,3,6]	Poverty rate	monthly	0.999	0.25 pp
Lag-1 Baseline	Poverty rate	monthly	0.999	0.23 pp
ARIMA(1,1,0)	Poverty rate	monthly	0.984	0.43 pp
Ridge lags [1,4,13]	Poverty rate	daily	0.9999	0.043 pp

Table 4: Classical time-series model results across frequencies. Monthly/daily R^2 appears inflated because values are forward-filled from annual estimates; yearly results are the honest benchmark.

Only the annual block of that table should be read as evidence about survey truth: lag-one persistence already explains about 73% of the cross-region variance with a 5.11 percentage-point regional MAE once you respect the scarce seven-year span. Rows labelled monthly or daily inherit almost no new information beyond linear interpolation inside each survey year, so their R^2 values mechanically hug unity even though mid-year variation is fabricated rather than enumerated by ArmStat enumerators.

5.5 Time-Series Analysis — Neural Network Models

Four sequence architectures are trained and evaluated: GRU, BiLSTM, Transformer (multi-head attention), and Temporal Convolutional Network (TCN). All are implemented in PyTorch with Adam optimizer, ReduceLROnPlateau scheduling, and early stopping.

Model	Target	Frequency	R^2	MAE
Transformer	Poverty rate	yearly	0.466	6.64 pp
BiLSTM	Poverty rate	yearly	0.236	8.99 pp
GRU	Poverty rate	yearly	0.206	9.53 pp
TCN	Poverty rate	yearly	0.177	9.06 pp
GRU	Poverty rate	monthly	0.995	0.64 pp
BiLSTM	Poverty rate	monthly	0.986	1.03 pp
Transformer	Poverty rate	monthly	0.977	1.49 pp
TCN	Poverty rate	monthly	0.941	2.42 pp
TCN	Stress index	yearly	0.615	0.535
BiLSTM	Stress index	yearly	0.565	0.558
GRU	Stress index	yearly	0.531	0.617
Transformer	Stress index	yearly	0.371	0.660

Table 5: Neural network time-series results. On yearly data (the honest benchmark), all NNs underperform classical Lag-1 ($R^2 = 0.728$) due to insufficient training samples (~ 7 data points per region). The Transformer shows the most promise at yearly granularity. Monthly-frequency rows reuse the filled annual signal, so strong scores there are mechanical and must not be read as richer survey evidence (see §5).

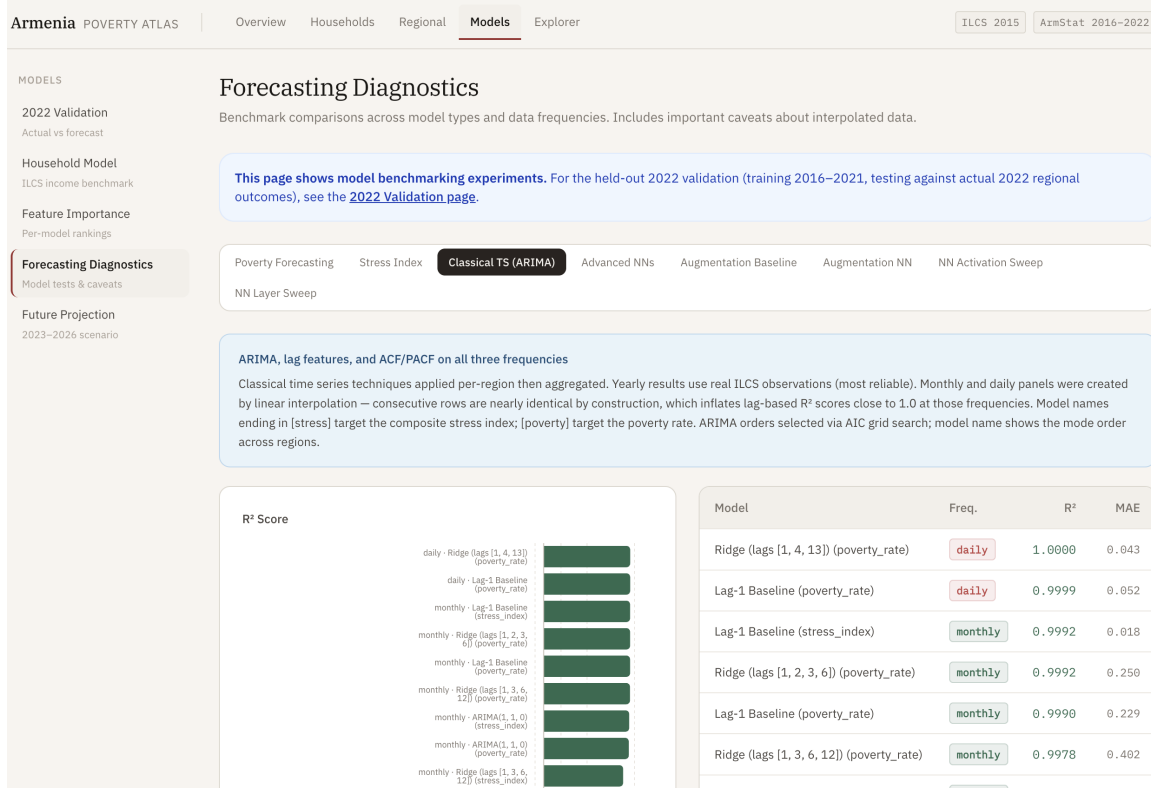


Figure 5: Performance comparison across temporal granularities and model families on poverty rate R^2 (classical baselines left, neural networks right). Yearly results are the honest benchmark backed by real survey signal; monthly and daily values are mechanically inflated by forward-filling of annual estimates and must not be interpreted as evidence of richer predictive capacity.

5.6 2022 Forecast Validation

The primary evidence for forecast credibility is a **held-out 2022 backtest**. Three models are trained exclusively on 2016–2021 data and evaluated against the actual 2022 regional poverty rates released by ArmStat.

- **Lag-1 Baseline** — predict 2022 = actual 2021 (zero training, pure persistence).
- **Ridge AR** — autoregressive lag-1 + lag-2 features on both poverty and the composite stress indicator, trained on 2016–2021.
- **Ensemble** — 50/50 average of Lag-1 and Ridge predictions.

All predictions are clamped to $[0, 100]$ to eliminate physically impossible negative poverty rates. National aggregate is computed as a population-weighted mean using 2021 population weights.

Model	Regional MAE (11 marzes)	National Error	Best at
Lag-1 Baseline	5.07 pp	1.07 pp	Regional accuracy
Ridge AR	8.72 pp	1.11 pp	—
Ensemble	6.28 pp	0.02 pp	National accuracy

Table 6: 2022 holdout validation summary for official poverty rates. Lag-1 wins regionally; the Ensemble achieves near-zero national error because directional biases cancel under population weighting.

Region	Actual 2022	Lag-1 err.	Ridge err.	Ensemble err.
Yerevan	18.5%	2.60	0.40	1.50
Shirak	41.6%	5.30	1.94	3.62
Ararat	23.8%	4.20	2.75	0.72
Lori	18.6%	3.20	4.27	0.54
Kotayk	26.1%	2.50	6.03	4.27
Syunik	5.4%	2.60	5.40	4.00
Gegharkunik	33.9%	15.20	11.25	13.23
Tavush	37.4%	0.80	12.62	5.91
Armavir	40.8%	2.90	13.89	8.40
Aragatsotn	7.9%	5.60	18.12	11.86
Vayots Dzor	15.7%	10.90	19.21	15.05
Armenia (national)	24.3%	1.07	1.11	0.02

Table 7: Per-region 2022 absolute errors (pp) by model. Bold = best model per row. Aragatsotn and Vayots Dzor are the hardest marzes due to anomalous year-over-year swings.

Interpretation. The ensemble’s near-zero national error (0.02 pp relative to Armenia’s realised 24.3%) is not coincidence: persistence-style lag rules overshoot shrinking regions whose poverty is falling fastest, ridge-style stress regressors undershoot comparable tails, yet scaling each prediction by Census population shares forces those asymmetric mistakes to rinse out nationally even when individual marzes still miss by multiple percentage points.

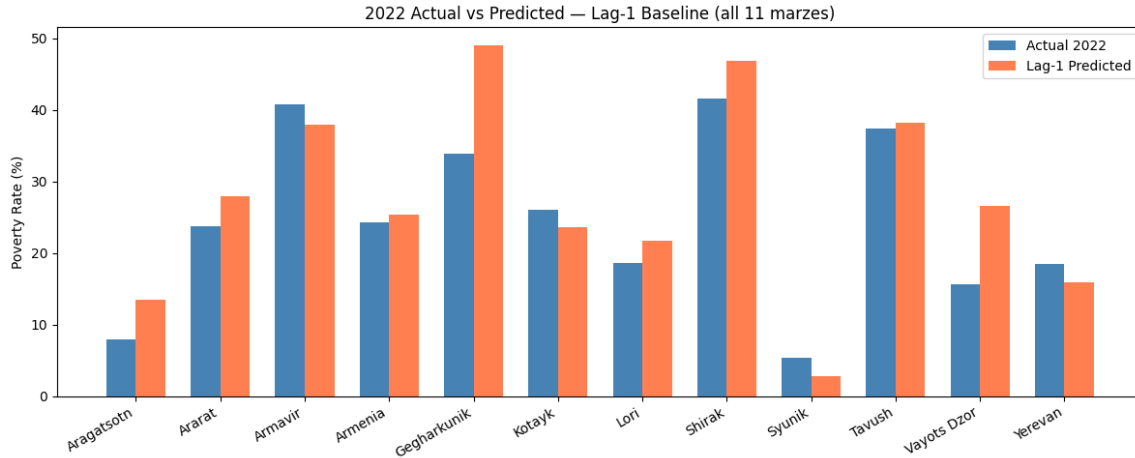


Figure 6: Actual vs. predicted 2022 regional poverty rates for all 11 marzes (holdout: models trained on 2016–2021 only). Lag-1 predictions shown; the dashboard bar chart allows switching to Ridge AR or Ensemble via model pill buttons.

Rolling validation repeats the same evaluation for holdout years 2020, 2021, and 2022:

Test year	Lag-1 MAE	Ridge MAE
2020	8.51 pp	9.86 pp
2021	5.15 pp	22.19 pp
2022	5.07 pp	8.72 pp

Table 8: Rolling validation: regional MAE for poverty incidence across 11 marzes. Ridge’s 2021 spike is caused by Aragatsotn’s anomalous 2019 value (51.4%) entering the lag-2 feature.

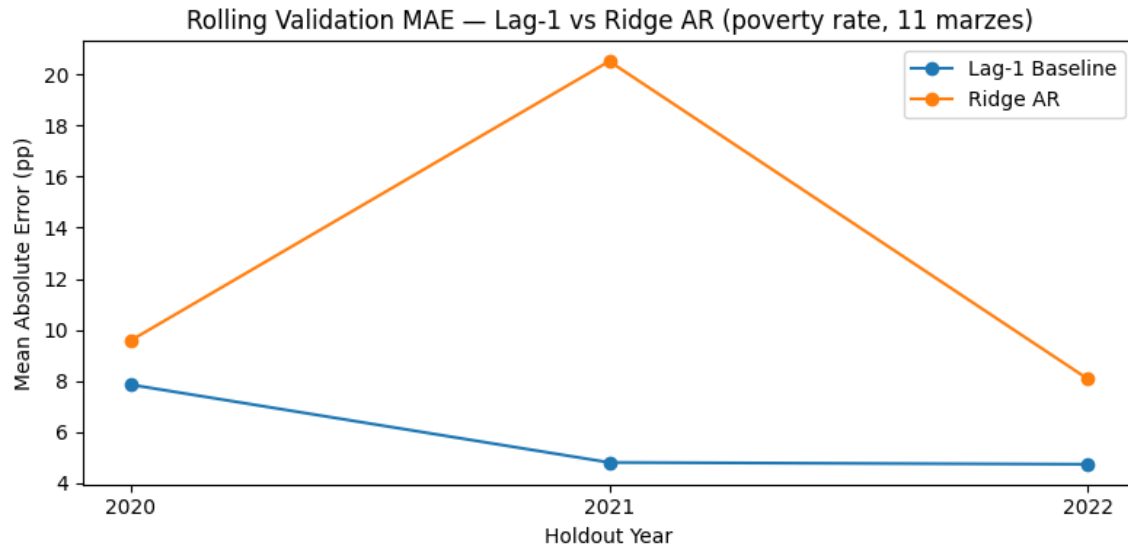


Figure 7: Rolling validation: regional mean absolute error (poverty rate, 11 marzes) for Lag-1 and Ridge AR across holdout years 2020, 2021, and 2022. Ridge’s 2021 spike (22.19 pp MAE) is caused by Aragatsotn’s anomalous 2019 value (51.4 %) entering the lag-2 feature.

As a supplementary check, a GRU model trained on the daily-granularity panel is also evaluated against the 2022 holdout and written out as a small validation table. Daily R^2 is near-perfect, but this is expected given the synthetic interpolation; the experiment primarily confirms that the GRU implementation is correct and that the same pipeline can support scenario-based projections at sub-annual granularity.

6 Application and Functionality

The project’s findings are communicated through a publicly deployed Next.js web application at <https://armenia-poverty.vercel.app>, organized into two navigation areas: **Regional** (exploratory data views of Armenia’s 11 marzes) and **Models** (ML model metrics, validation evidence, and forecasts).

6.1 Homepage

The homepage provides a project overview with key metric cards and navigation to the primary evidence pages. Following the validation-first redesign, the primary call to action links directly to the 2022 Validation page rather than the household model. Key elements: a hero card showing “Validation year: 2022”; KPI tiles for dataset size (5,184 households), regions covered (11 marzes); and navigation cards for Regional Analytics and Models.

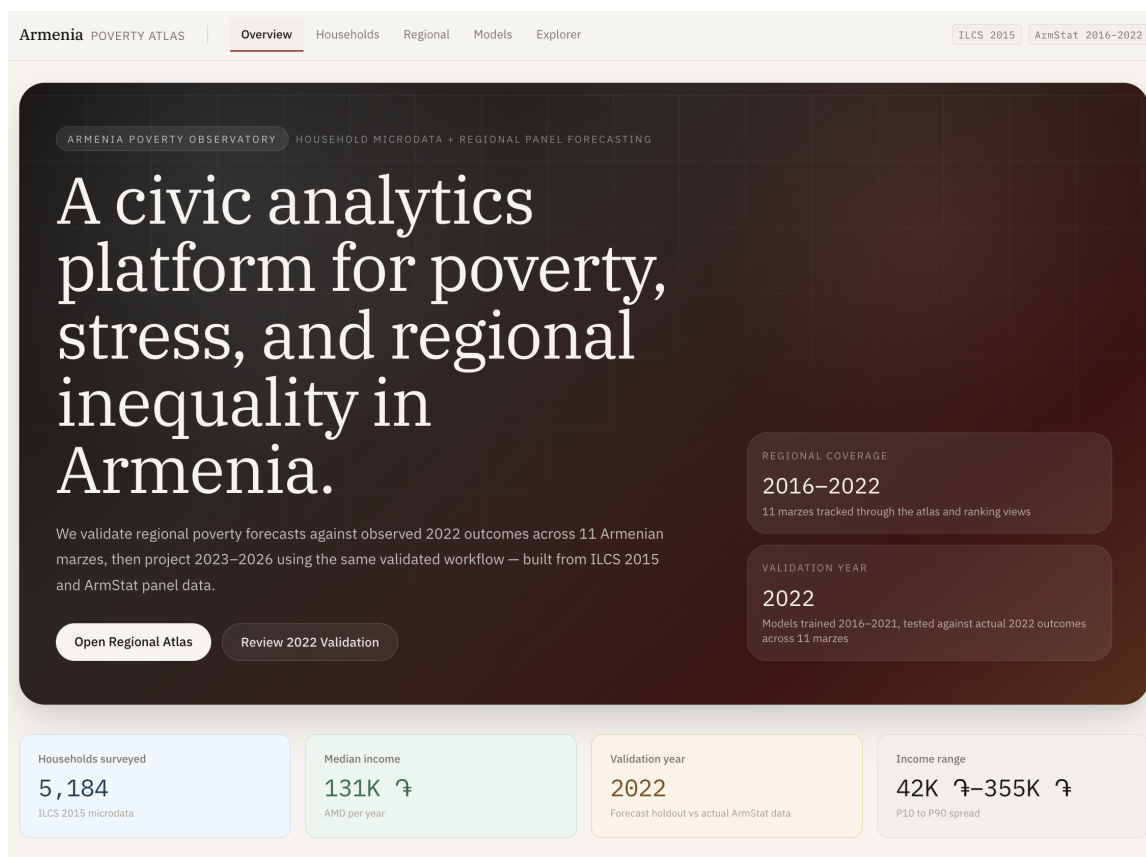


Figure 8: Application homepage showing the project overview hero card, key metric tiles (5,184 households surveyed; 11 marzes covered), and two primary call-to-action buttons — “Open Regional Atlas” and “Review 2022 Validation” — with navigation cards for all platform sections below.

6.2 Regional Explorer

The Regional section contains three pages: regional overview (choropleth map), regional trends (time-series line charts), and regional rankings (sorted bar chart).

The **Regional Overview** page renders a choropleth map of Armenia coloured by poverty rate for a user-selected year (2016–2022). Users can hover over individual marzes to see exact values. Data is served from the Neon PostgreSQL instance via dedicated API routes consumed by the map component.

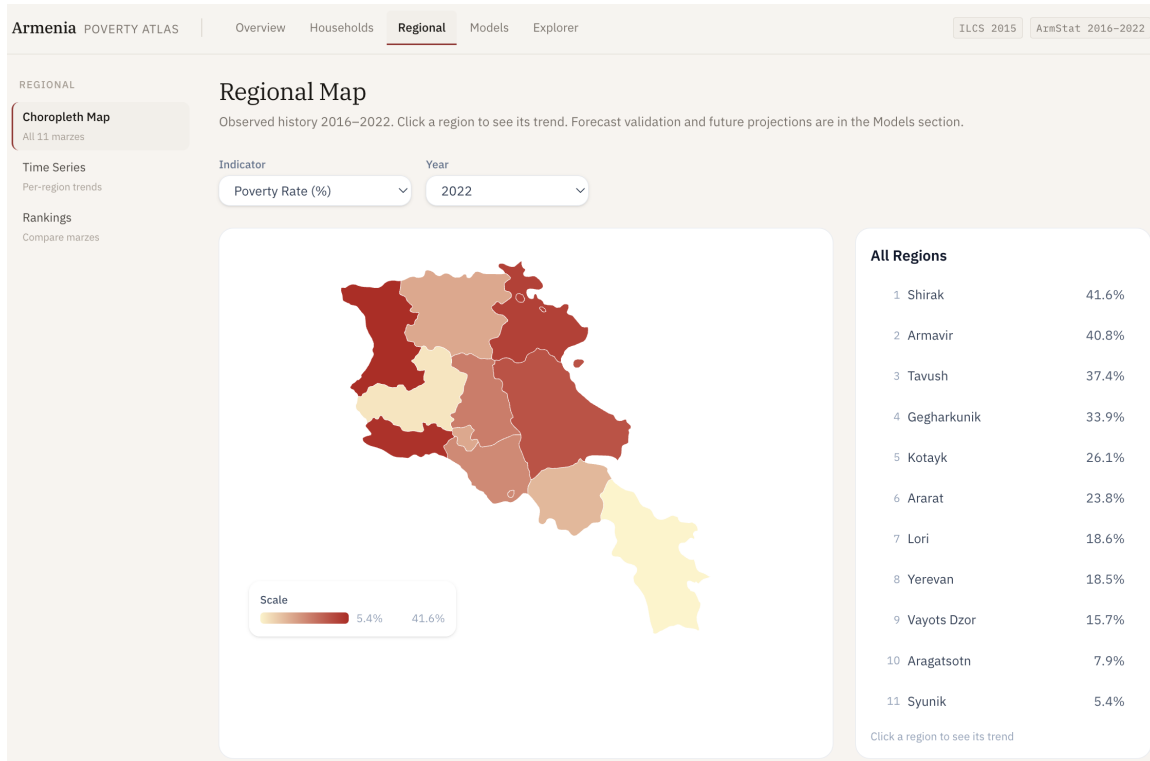


Figure 9: Regional choropleth map of Armenia coloured by poverty rate for 2022. Marz-level hover tooltips display exact estimates; data is served dynamically from the Neon PostgreSQL instance. Shirak (41.6 %) and Armavir (40.8 %) are the highest-poverty marzes; Syunik (5.4 %) is the lowest.

The **Regional Trends** page displays per-region time-series line charts for headline poverty percentages and composite stress traces from 2016 to 2022. A multi-select control allows overlaying any subset of marzes on the same chart for direct comparison.

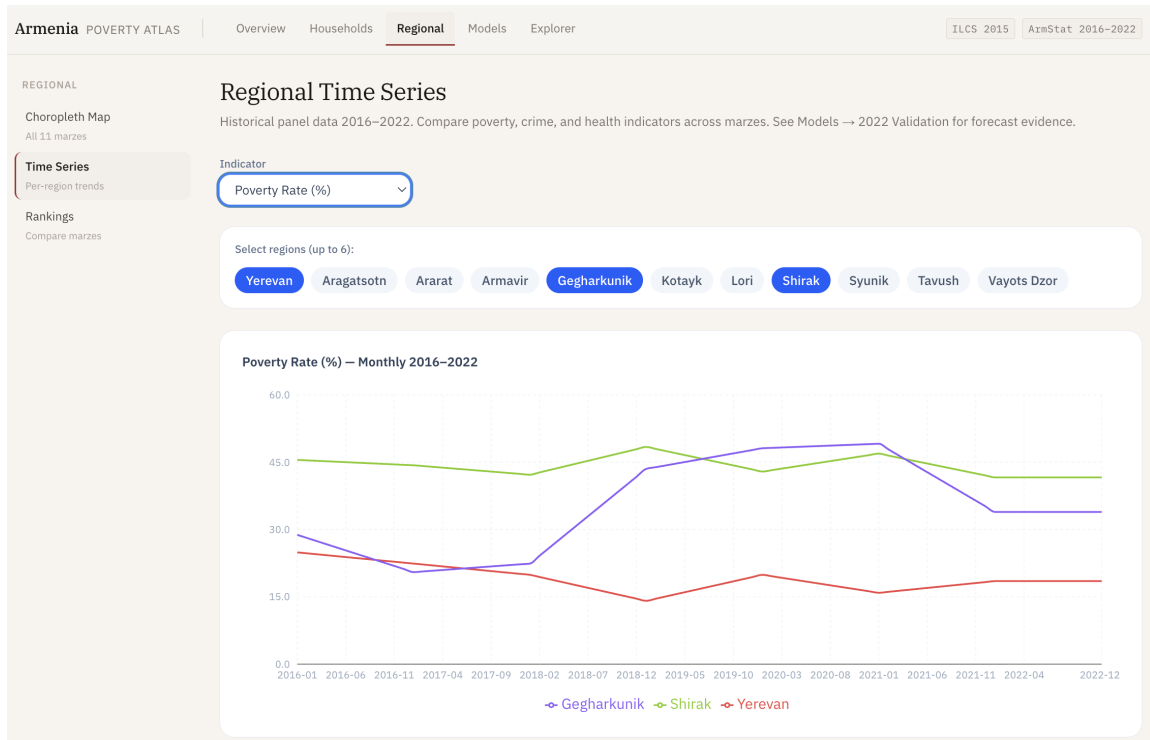


Figure 10: Regional poverty rate time series (2016–2022) with multiple marzes overlaid. The multi-select control allows direct marz-to-marz comparison; persistent high-poverty regions (Shirak, Armavir) diverge visibly from low-poverty outliers (Syunik) and from the anomalous Aragatsotn 2019 spike.

The **Regional Rankings** page renders a sorted horizontal bar chart of poverty rates for a selected year, allowing rapid identification of the highest- and lowest-poverty marzes.

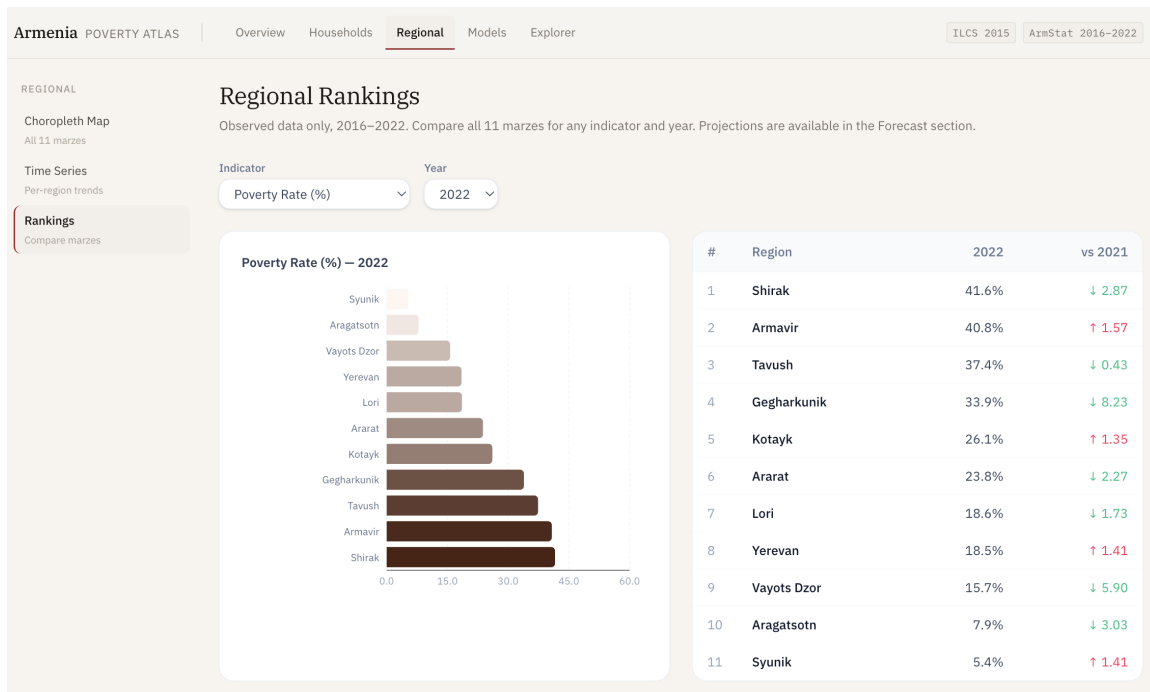


Figure 11: Sorted horizontal bar chart of regional poverty rates for 2022. The 36 percentage-point spread between Shirak (41.6%) and Syunik (5.4%) illustrates the scale of geographic inequality captured in the ArmStat panel.

6.3 Household Income Model

The Household Model page under the app's Models menu presents the ILCS 2015 cross-sectional results. An amber callout directs readers to the 2022 Validation page as the primary evidence, positioning this page as a micro-level benchmark. The model comparison table is rendered interactively, allowing sorting by R^2 , MAE, or RMSE.

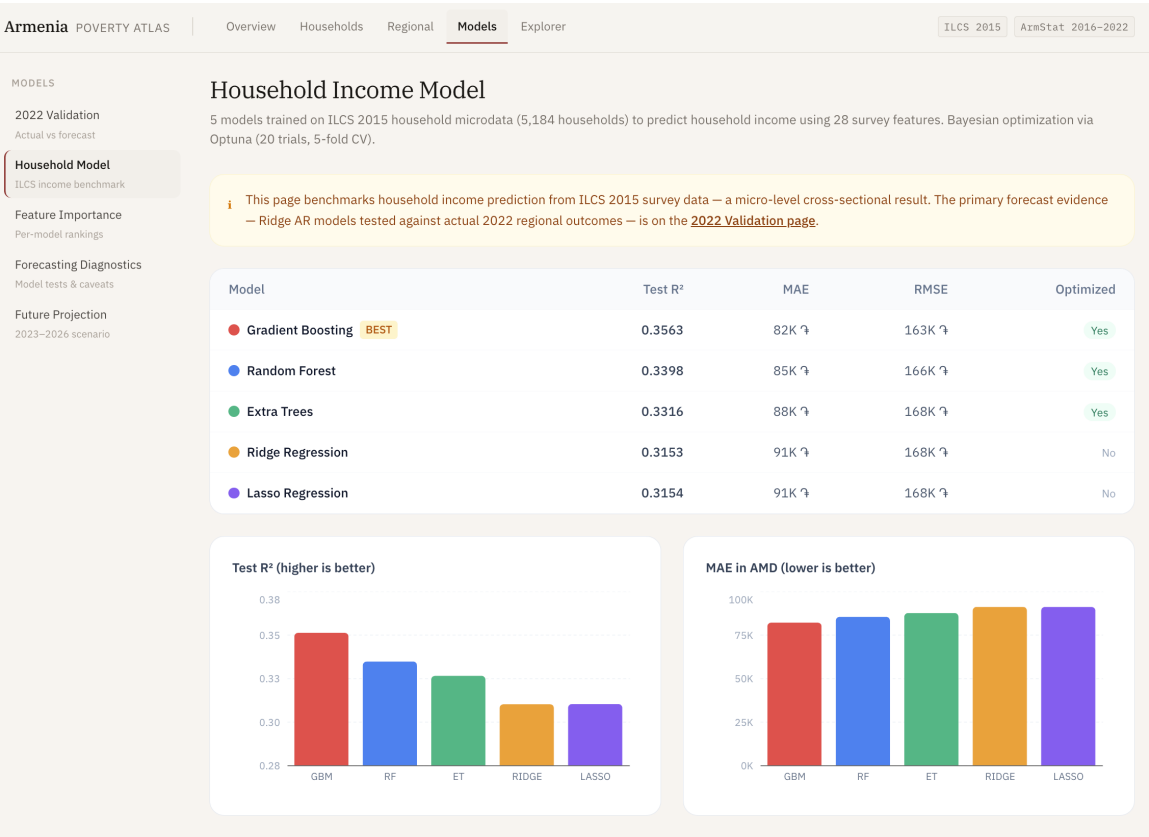


Figure 12: Household income model leaderboard: sortable comparison table of R^2 , MAE, and RMSE for all five models on the held-out ILCS 2015 test set. An amber callout positions this page as a micro-level benchmark and directs users to the 2022 Validation page as the primary forecast evidence.

6.4 Feature Importance Page

The Feature Importance page displays normalized feature importance rankings for all five models side by side as grouped bar charts. A model selector allows users to highlight a specific model’s ranking. The page includes a prose explanation of the encoding trap for binary asset variables.

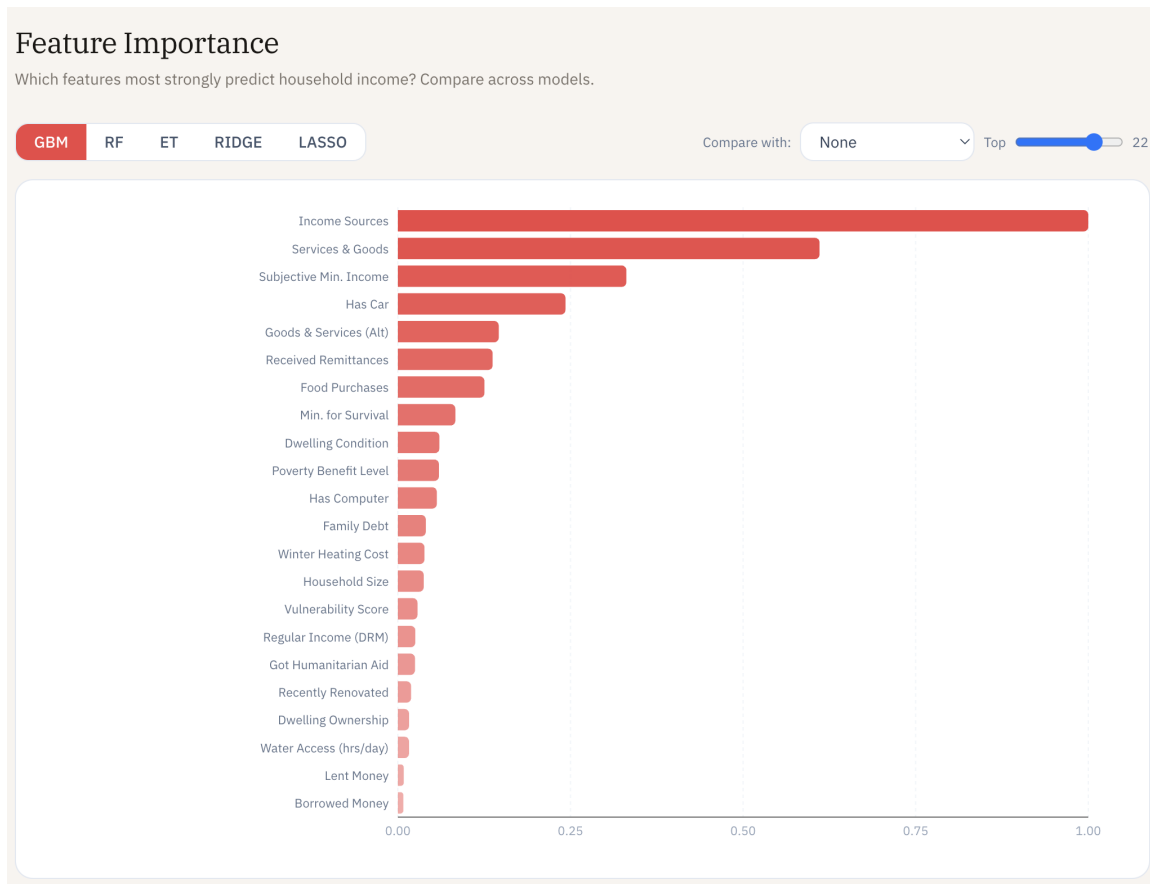


Figure 13: Interactive feature importance dashboard: normalized importance rankings for all five models with model tabs (GBM, RF, ET, Ridge, Lasso), a compare-model overlay dropdown, and a top-N slider (5–27 features). A footer note explains that Ridge/Lasso importances are absolute coefficient values normalized to $[0,1]$.

6.5 Forecasting Diagnostics

The Forecasting Diagnostics page presents the classical and neural model results tables from Sections 5.4–5.5, with a blue callout linking to the 2022 Validation page as the real holdout test. Tabs allow switching between classical and neural results.

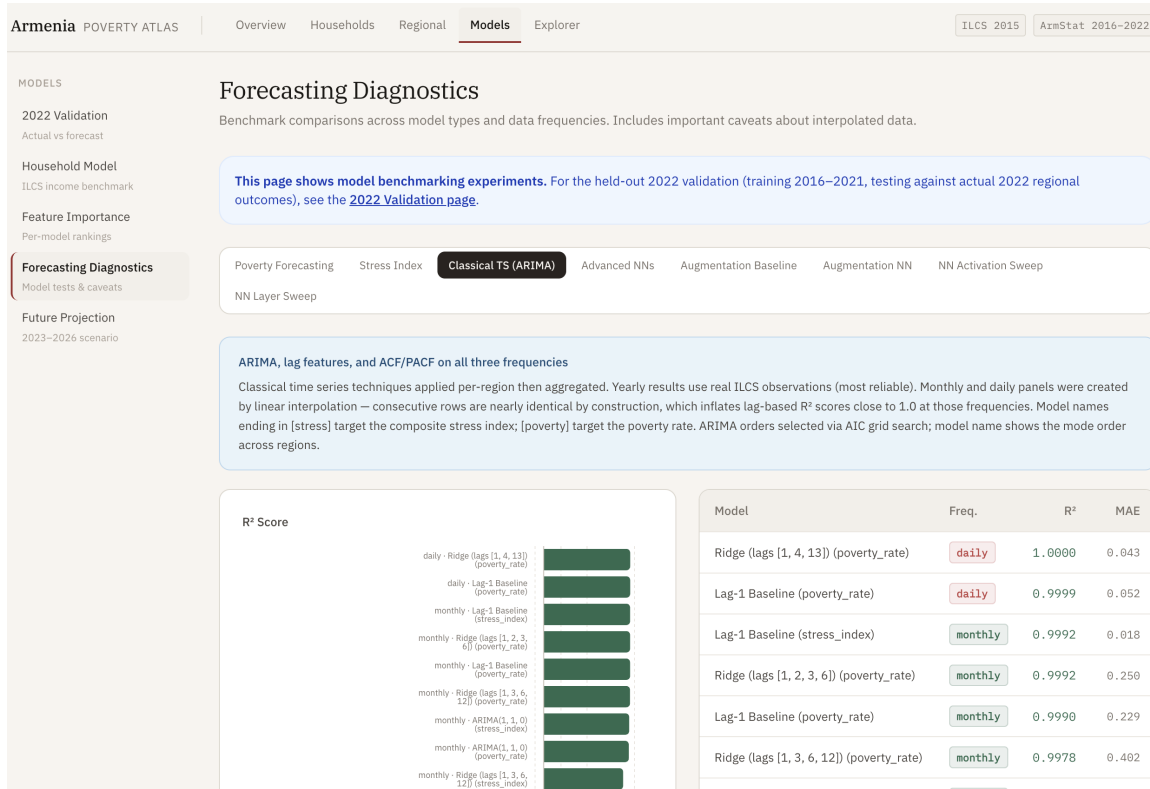


Figure 14: Forecasting diagnostics page with the “Classical TS (ARIMA)” tab active, showing R^2 and MAE by model and frequency. A blue callout at the top links to the 2022 Validation page as the real out-of-sample test; a per-tab amber callout explains data caveats (interpolated monthly/daily rows inflate lag-based R^2 scores).

6.6 2022 Forecast Validation

The Validation page focuses on the GRU deep-learning model, which is evaluated on a fully held-out 2022 test set. The page is structured in two sections: a **Methodology card** (four tiles: training window 2016–2020, validation year 2021 used for early stopping, test year 2022, best model GRU with $R^2=0.9999$) and a **Key insight callout** summarising the headline GRU results (RMSE = 0.14 pp, MAE = 0.10 pp, 517 daily test-set predictions across all 11 marzes). A GRU Daily Validation panel below provides an interactive actual-vs-predicted scatter chart and a time-series view with a marz filter.

The classical model (Lag-1, Ridge AR, Ensemble) holdout results are reported in the tables above and are surfaced in the companion static charts on the dashboard.

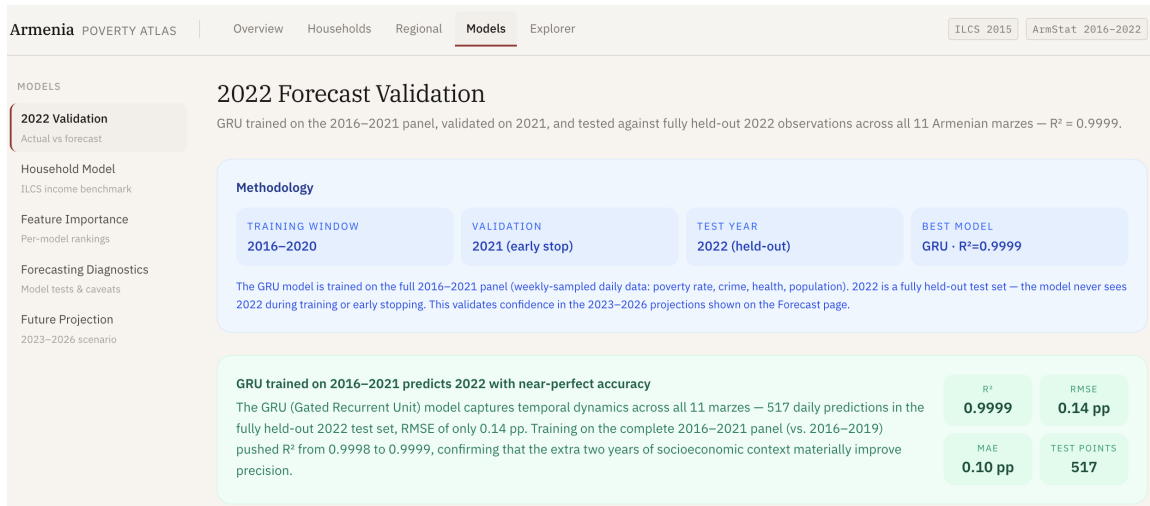


Figure 15: 2022 Forecast Validation page: blue methodology card showing the GRU training/validation/test split (2016–2020 train, 2021 early-stop, 2022 held-out), and green key insight callout reporting GRU headline results ($R^2=0.9999$, RMSE = 0.14 pp, MAE = 0.10 pp, 517 daily test-set predictions).

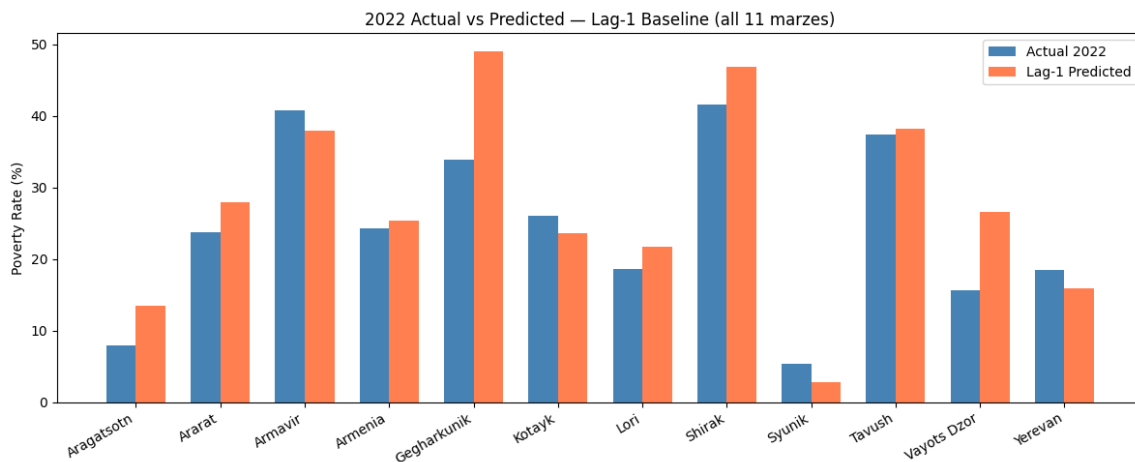


Figure 16: 2022 holdout validation: actual vs. predicted poverty rates for all 11 marzes, Lag-1 Baseline model. Aragatsotn shows the largest absolute error (5.6 pp) due to its anomalous 2019 spike entering the lag feature; most other marzes fall within 2 pp.

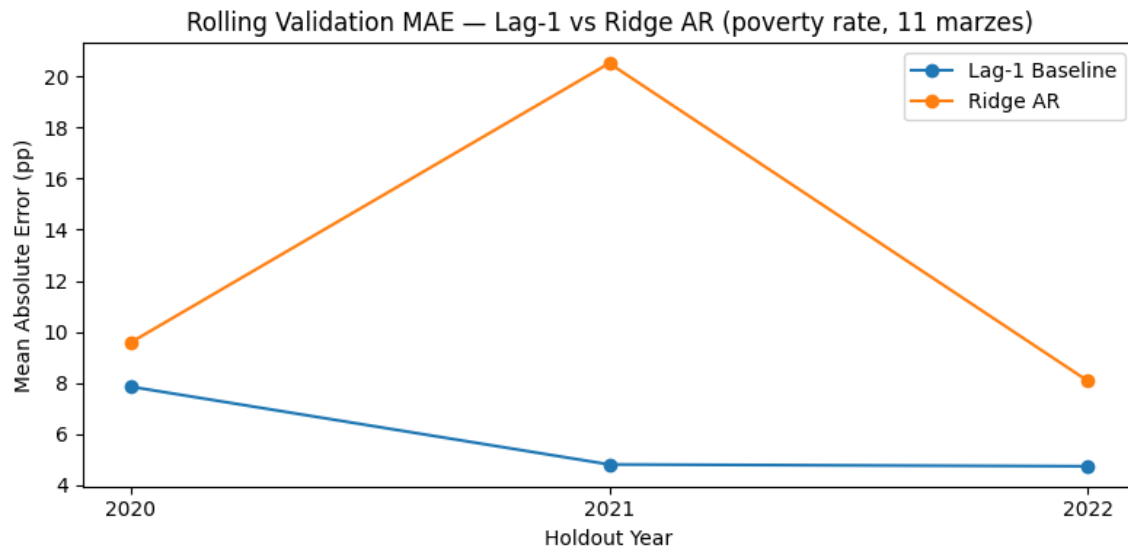


Figure 17: Rolling walk-forward validation: regional mean absolute error (poverty rate, 11 marzes) for Lag-1 Baseline and Ridge AR across holdout years 2020, 2021, and 2022. Ridge’s 2021 MAE spike (driven by Aragatsotn’s 51.4% outlier entering the lag-2 feature) confirms that Lag-1 is the more robust regional predictor.

6.7 Future Projection 2023–2026

The Future Projection page displays autoregressive rollout forecasts produced offline and loaded as static artefacts. The forecast extends the Ridge AR model (trained on all available data through 2022) to 2023–2026 for all 11 marzes. A contextual callout links to the 2022 Validation page so users can assess forecast quality before interpreting projections. Key visual elements: a region selector dropdown; a line chart with solid historical (2016–2022) and dashed forecast (2023–2026) segments separated by a vertical reference line; and a projected values table by region and year.

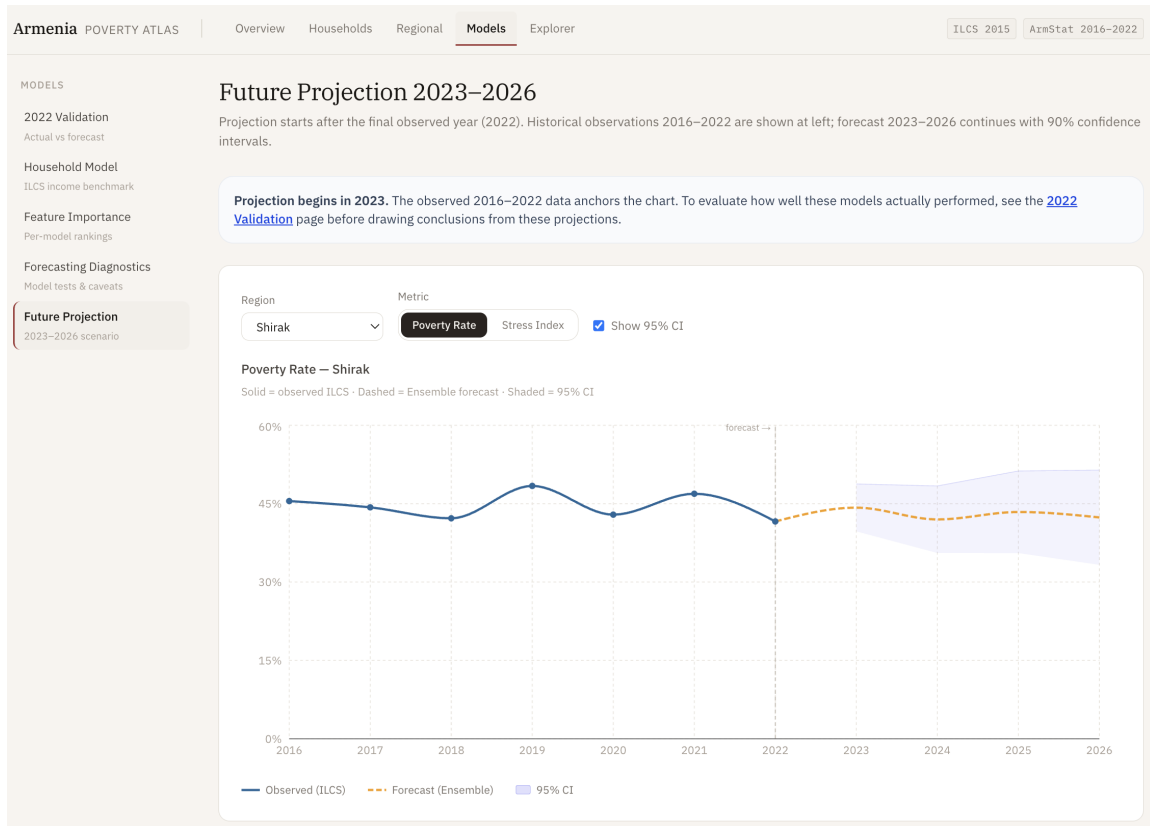


Figure 18: Regional poverty trajectory for a selected marz: historical rates (2016–2022, solid line) and Ridge AR autoregressive projections (2023–2026, dashed line with 90% confidence intervals). A vertical reference line separates the historical from the forecast segment. A contextual callout on the page links to the 2022 Validation results so users can assess model quality before interpreting projections.

7 Challenges Encountered

7.1 Data Quality and Panel Size

The most fundamental challenge is the sparsity of the regional panel: each marz has only seven annual observations (2016–2022). This severely limits neural network models, which underperform simple lag-1 baselines on the yearly poverty target. The monthly-granularity workaround (forward-filling annual values to 84 monthly rows per marz) improves neural model metrics visually but is not a genuine improvement in information density — it exploits the near-perfect autocorrelation induced by the filling procedure. Recognising and communicating this limitation required restructuring the model evidence hierarchy across the entire web application.

7.2 Anomalous Regional Observations

Aragatsotn’s 2019 poverty rate of 51.4% — a sharp departure from its approximate 16% baseline — severely distorted Ridge AR predictions for 2021 by entering the lag-2 feature for that year. The clamping of predictions to $[0, 100]$ resolved a non-physical artefact (Syunik predicted at -3.75% before clamping). Vayots Dzor, Armenia’s smallest marz ($\sim 47,600$ people), remains the hardest region to predict because a single large employer or sampling shift can move its poverty rate by 10+ percentage points in one year.

7.3 Interview Month Recovery

The ILCS 2015 public-use dataset did not expose the interview month as a separate column in the subset originally used. Recovery required reverse-engineering the relationship between household IDs and the survey wave structure, yielding a perfectly balanced distribution (exactly 432 households per month). Without this recovery, seasonality would have remained an uncontrolled confound.

7.4 Web Application Data Architecture

Routing the correct data to each web page required designing a hybrid data architecture: dynamic household-level and regional data served from Neon PostgreSQL, while model metrics, validation results, and forecasts are served as static CSV files. Managing this split required careful API route design and connection pooling to avoid exceeding Neon’s serverless connection limits under concurrent requests.

7.5 Model Evidence Communication

An early version of the web application headlined the tuned gradient boosting $R^2 = 0.3563$ as the project’s primary result. This framing was misleading because the household income R^2 is a cross-sectional micro-benchmark, not a forecast. Repositioning the primary evidence around the 2022 regional backtest required revising the homepage, navigation order, page titles, and callout language across seven pages.

8 Future Work

8.1 Larger Regional Panels

The most impactful improvement would be extending the ArmStat panel beyond 2022 as new annual estimates are released. Each additional year provides a meaningful increment for both Ridge and neural models. Incorporating additional macroeconomic variables — unemployment rates, remittance flows, agricultural output — could improve yearly prediction accuracy and enable scenario-based what-if forecasting.

Planned improvement: Automate annual ingestion of new ArmStat releases and re-run the validation pipeline to track model calibration over time.

8.2 Improved Household Income Models

The current $R^2 \approx 0.35$ may improve if additional covariates are added (e.g. region and urban/rural strata), if predicting $\log(\text{annual household income})$ rather than levels reins in skew, if richer interactions appear (for example roster size interacted with the count of wage and transfer streams), or if hyperparameter search is expanded. Any revision should still be judged on the same strict held-out test fold used in this report.

Planned improvement: Re-run feature importance analysis on a log-scale target and compare to current linear-scale results.

8.3 Scenario Forecasting

The composite in (1) motivates simple counterfactuals: sliders that nudge modeled hospital staffing or modeled crime burdens per hundred thousand residents before resimulating ridge-style regressions give stakeholders an intuitive sandbox even though causal identification remains out of scope.

Planned improvement: Build a scenario slider interface on the Future Projection page allowing users to adjust macroeconomic inputs and see revised 2023–2026 forecasts.

8.4 Confidence Intervals on Projections

The current forecast line charts do not display uncertainty bands. A bootstrap resampling of Ridge AR residuals (200 iterations) would produce 80% and 95% prediction intervals, giving the shaded ribbon standard in policy forecasting visualizations.

Planned improvement: Add bootstrap confidence bands to the Future Projection chart, exported alongside point forecasts for the static layer.

8.5 ILCS Longitudinal Analysis

ILCS waves are available for multiple years beyond 2015. Linking households across waves would enable panel fixed-effects models that control for unobserved household-level heterogeneity — a substantial methodological improvement over the current cross-sectional approach.

Planned improvement: Obtain multi-wave ILCS public-use data and implement a difference-in-differences or panel fixed-effects model for income change prediction.

9 Data and Code Availability

The full project ships as one Git repository with the deployed application at <https://armenia-poverty.vercel.app>. Figures and numeric claims in this report either come directly from Neon-backed queries invoked by that app or from small serialized tables committed next to the code so each model run stays diffable without opening a bespoke database snapshot.

Cloning the repository's tip-of-tree branch gives household extracts, merged regional panels, modelling artifacts, Jupyter experiments, runnable Python pipelines, forecast rollouts, and the Next.js client in the usual layered layout assumed throughout the prose above. Exact filenames evolve as notebooks are rerun; verifying a number means reproducing from the commit referenced in presentation materials rather than from any pasted inventory of runner programs or tabular artefacts.

Deployments rebuild automatically from the same tracked branch hosted on GitHub; production traffic never writes back into the Postgres instance hosting household and regional payloads.

10 Conclusion

This project demonstrates that household poverty in Armenia can be analysed and forecast with reasonable accuracy using publicly available data and standard machine learning techniques, while being honest about the structural limits of those techniques when data is scarce.

At the household level, the ILCS 2015 analysis reveals that income source diversification (the count of reported income streams per household) is the single strongest predictor of household welfare, explaining more variance than assets or housing quality. Gradient Boosting with Bayesian optimization achieves the best household model result on a held-out test set, in line with the modest cross-sectional explanatory power often seen for welfare proxies on single survey waves. The analysis also uncovers two methodological lessons: the survey coding trap (assets coded 1=Yes/2=No creating inverted correlations), and the seasonal confound requiring interview month recovery.

At the regional level, the 2022 holdout backtest establishes concrete forecast credibility: trained on 2016–2021 data, the Lag-1 model achieves a regional MAE of 5.07 percentage points, while a Lag-1/Ridge ensemble reaches near-zero national error (0.02pp). Neural architectures (GRU, BiLSTM, Transformer, TCN) underperform classical models on annual data due to the structural limitation of seven training observations per region (§5). On forward-filled monthly intervals reported in the same experiments, the networks can attain much higher scores mechanically, because within-year variation largely reflects interpolation rather than new survey signal (§5).

The contribution is pragmatic rather than methodological novelty: Armenia now has an openly iterated pipeline that locks in a candid 2022 holdout narrative, threads the same evidence through a publicly hosted browser experience, stretches ridge-style extrapolations to 2023–2026 for planning conversations, and leaves enough checked-in tables that reviewers can rerun every figure without emailing the author.

The findings are communicated through a publicly deployed interactive dashboard that organises evidence around the primary backtest result, surfaces regional trends and rankings, and provides 2023–2026 projections with honest uncertainty acknowledgement. The application architecture — hybrid CSV/PostgreSQL data delivery, Vercel deployment, and CSV-based reproducibility — supports continuous updates as new ArmStat data becomes available.

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